Syllabus MS&E 152 Summer 2026

** Introduction to Decision Analysis** · Stanford University · 4 units

Course Links

- Canvas
- Ed Discussion
- Course Website

Course Description

Decision Analysis is an application of Decision Theory to making good decisions when outcomes are uncertain. It applies to practical problems in engineering, organizations, and even one's personal life. It is a principled, quantitative approach that draws upon theory in Economics, Applied Probability, and Psychology. These principles of rational choice extend to applications that combine individual judgment with quantitative, data-driven modeling techniques.

This course will develop skills in structuring and solving problems through interactive and project class work. These methods can be applied to assisting an individual, a group, or an organization making a decision. By working with probability and value models, students will gain an understanding how to quantify belief, express risk and time preference, and how information can be valued. An introduction to Influence Diagrams will address how decisions are affected by how a problem is framed. This will be illustrated by cases from finance, project management, law, and medicine. In conclusion we will touch on issues of Decision Quality, the limits of rationality, and the ethical implications of a decision.

Prerequisites

As this is an Engineering course, ability in high-school level mathematics and spreadsheets is needed. A background in probability is helpful. Co-registration in an introductory probability course is appropriate for students with no previous study. Basic software or programming (e.g. python notebooks) experience is entirely optional.

Pre-course assignment

- Review your understanding of basic probability with the probability refresher.
- Watch this TED talk on the value of improving personal decision making.

Course Learning Objectives

By virtue of taking this course you should:

1. Understand how the fundamentals of rational choice lead to the definition of a good decision, as distinct from its eventual outcome.
2. Be able to apply the probability "calculus" by which belief can be elicited and uncertainty quantified,
3. Know how to formulate a decision model that integrates decisions, probabilities and values,

4. Be able to compute how a decision model can determine whether additional information is worth acquiring.
5. Be skilled in how one can assist others by the process of Decision Analysis and application of the principles of Decision Quality.

Teaching Team

Role	Name	Contacts	Location
Instructor	John Mark Agosta, Ph.D.	jmagosta@stanford.edu 650 465-4707	Huang 261
Teaching Assistant	Uluk Batyrgaliev	ubtrglv@stanford.edu	Zoom

Class Hours

The course runs for eight weeks (Jun 22 – Aug 13, 2026), with two lecture sessions and one disscussion section per week.

- **Lectures:** Mon/Wed, 10:30 AM–12:30 PM · Y2E2 111
- **Discussion Section:** Thu, 3:00–4:15 PM · Y2E2 111
- **Office Hours:** On days following lectures, after lunchtime.

Lectures and discussion sections will not be recorded. Lecture notes will be posted on the course website after each session.

Discussion and Office Hours The best way to communicate is in person, after lecture or during office hours. Real-time meetings are more efficient, more effective, and more fun. Discussion sections will offer of help with the course material. Be sure to bring your laptop! You will need to attend some discussion sessions where you will be asked to explain how you solved one of your previous homework problems.

Duration of office hours will depend on attendance and demand. You can assume that there will be sufficient opportunities to meet the needs of your schedule. We will also make ourselves available to meet with students by appointment.

Online Discussion

All online forum discussions will take place using **Ed Discussion Platform**. Please use it as the exclusive online tool to ask questions, discuss topics of common interest, and to create and run your project teams. We encourage you to ask questions whenever you want help to understanding material in the course. We will use the Ed Discussion Platform exclusively for online discussions. "Third channel" media that do not include all students are unfair to those not included.

Textbooks and Other Materials

As with any college level course, you will be expected to read the required course material for each lecture before attending the class. The class schedule shows the assignments of videos, required and optional readings for each class.

- **Course Notes (primary)😗*The textbook for the course consists of excerpts, with revisions, from Professor Howard's draft manuscript, Foundations of Decision Analysis

- **Optional Textbook:** Students desiring a comprehensive source on Decision Analysis can instead purchase the book of the same title: R. Howard and A. Abbas, (2016) **Foundations of Decision Analysis**, Pearson. It can also be found among the books on reserve for the class in the Terman Engineering Library.

- **Videos and Supplemental Readings:** There are video links to `youtube` recordings that are an gentle way to familiarize yourself with class content. For those interested in pursuing subjects in depth there will be optional further readings of interest distributed online, and on library reserve.

- **Software:** Students should be familiar with the use of Canvas, as the "top level" portal for the course. We will use it for announcements, links to other sites, as well as for homework submission and tracking grades. However all class material such as readings and lecture notes, homework guides will be distributed as files, downloadable on the class website.

Occasionally we may distribute via the class website python notebooks or scripts to illustrate points made in lecture.

For examples and demonstrations we will use the *Genie* graphical Bayes network editing and solution tool available for download from *BayesFusion* without charge for academic use. It is available optionally for use in class projects. It is entirely possible to meet the requirements for the class project without use of software tools.

Course Topics and Fulfillment of Requirements

The principal objective of the course is to make you comfortable understanding and using concepts and tools fundamental to the application of Decision Theory. The course's subject matter is accessible to a wide range of interested parties, from students at a high-school level who are attending Stanford, to undergraduates exploring fields for a major, to graduates taking courses outside their department, and to "non-matriculates" who are not currently in a degree program and wish to tune up their knowledge of the field.

The course's content is not technically challenging, but the concepts around decision making are subtle, and understanding what makes a good decision require careful thought. It sharpens one's thinking by focusing on the logical progression of ideas, and promises to change how you will approach problems. When discussing what to do, our common language is not precise enough to express the nuances of probability, information, options, cause, preference, and risk. Decision Analysis is a means to bridge the gap between "hard" data and "subjective" judgment consistently, resulting in clarity of action.

As for background about the origins of the course, Decision Analysis has been taught at Stanford for over half a century, has fostered an extensive discipline of professional consulting, and has influenced the development of automated reasoning and AI.

MS&E 152 is offered for 4 units. For undergraduates the course fulfills the Ways of Thinking / Ways of Doing in either:

- *Formal Reasoning* (understanding fundamental concepts and distinctions about probabilities, decisions, and values under uncertainty) or

- *Applied Quantitative Reasoning* (applying a structured approach to making important personal and professional decisions).

We realize that there will a wide range of backgrounds and education levels in a summer quarter class. For those wishing enrichment and challenges matching their abilities we have included a detailed set of readings from the optional textbook, and from the primary literature. Office hours are a good opportunity to explore topics at greater depth. Group projects are another opportunity for exploration of advanced topics. Students wishing to pursue advanced projects including software development are encouraged to discuss them with us.

Attendance

Lectures will be in person only. Class participation is important. Since there are interactive class activities students are expected to attend class sessions to the greatest extent possible. Please **DO NOT** register for this course if you have a recurring conflict with the lecture times. Attending in person is encouraged through our participation policy.

Course Grading

The course grading scheme is designed to encourage students to keep up with
the course content in a timely manner, and to participate in class sessions in person to the extent possible. Please **DO NOT** register for this course if you have a recurring conflict with the lecture times.

Component	Weight
7 Homeworks (2 pts each)	14%
15 Quizzes (2 pts each)	30%
Midterm	16%
Group Project	30%
Class Participation & Attendance	10%

Grading Exceptions

Re-assigning points We recognize that students are balancing many priorities, and so we make accommodations among assignments for missing homework and quizzes. Students may assign points from missed homeworks or quiz assignments to other assignments. For example, students who are unable to attend a quiz or complete a homework may choose to increase the percentage assigned the midterm exam by 2 points. Please email us with your re-assignment choice no later than when the assignment is due. Midterm and project assignment points may not be re-assigned.

Slip days. To accommodate late homework, students have three "slip days" that may be used through the quarter with no grade penalty. For example, you could submit one homework two days late and a third one day late with no penalty.

Accommodations

If you have a letter from the *Office of Accessible Education* that grants extra time or modified test-taking conditions, fill out the form with your letter attached. To get started, or to initiate services if you do not have such a letter, please visit oae.stanford.edu. Other extensions on assignments will be granted only with an academic accommodation letter from the Office of Accessible Education, generally for medical reasons, or in other such exceptional circumstances. Requests due to job interviews, athletic competitions, other classes and assignments, and poor planning will not be considered.

Honor Code

The Honor Code is an agreement between students and faculty to create a productive, trusting, and fair learning environment. It is taken seriously at Stanford University, to be observed by both students and the teaching team. Simply put, we all share it places the responsibility for ensuring honest behavior on students as well as the teaching team, and violations should not be tolerated.

Respecting individual academic freedom as a prerequisite for learning, and the classroom is meant as a safe space, where honest opinions from all parties are are accepted equally and treated with mutual respect. It is our intent that students from all backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength, and benefit.

Assignments and exams

- **Homework** The goal of the homework is to help you practice the skills that you'll use for other assignments in this class and - we hope! - later in life. Please take the opportunity to understand it and think deeply about it. You will lack the ability needed for other assignments if you just submit answers that you look up online or get from AI tools without fully comprehending the homeworks.

Homework will be assigned weekly on Wednesdays, due the following Monday. It will be graded on a completion basis: You will receive a full score if you made an honest effort to complete it. Partial credit is assigned for incomplete submissions.

Discussion sections are your opportunity to inquire about the homework assigned for that week. Our policy is geared to make sure you can get the help you need, and so that by the time you turn in your work you understand what it's about, how it works, and why it's important. In addition, you will be asked to come to office hours at least twice during the quarter and to review your homework answers verbally, one on one.

Collaboration. Recognizing that in "real life" the best work is done in teams, collaboration on homework is encouraged. However, the homework you submit must be your own work, not copied from your collaborators, or from other sources. Under the Stanford honor code, students are allowed to collaborate with each other and with AI on homework. We ask that you --

- Give credit to the people who have helped you: please write on your homework the names of your study group -- the people you worked with.
- Give credit to the other resources that have helped you: please write on your homework your sources: the texts, notes, web pages, or AI models and details of how you found it useful. For example include a summary of your prompt and the response an AI returned you.
- *Write up your homework by yourself. That is, all of the text that you submit should be typed or hand-written by you.*

- Submit your homework as a file on Canvas. We will read and review all homework submissions and return them to you.

Posting solutions. Under no circumstance should you seek out or look at solutions to assignments given in previous years, or share or post solutions (yours or ours) in a way they are accessible outside of the class.

- **Quizzes**. Each class will begin with few minutes set aside for a one question "pop" quiz covering the previous course material. We will be able to record class attendance via quizzes and occasional in class polls (via *Poll Everywhere*).

- There will be **one in-class mid-term exam**, using novel probability-based scoring that draws upon concepts from the course. You are allowed a single page hand-written reference sheet to take with you to the exam.

Quizzes and exams will be written and turned in on paper. They will be closed book, without computer or other electronic devices.

- The **final group project** to formulate and analyze a decision analysis is mandatory. You will propose your group, consisting of 2 to 4 students. Your group is free to explore a situation that can be modeled as a decision, possibly for a decision-maker of your choice. We will post detailed instruction and guidance for coming up with projects on the class website. Each group will give a short presentation of their projects in the last section meeting, which you are expected to attend. Projects presentations will be submitted, e.g. as a slide deck for final grading. Projects will be graded by their adherence to the six principles of Decision Quality.

We will make available various software options for creating decision models, however use of software is not mandatory: It is possible to complete a decision analysis entirely on pencil and paper.

- **In class participation** will be determined by a combination of attendance, voluntary participation in class and session discussions, and other contributions made to the course.